

Semester	S.E. Semester IV – CMPN
Subject	R-Programming
Subject Professor In-charge	Prof. Sanjeev D.
Assisting Teachers	Prof. Prakash P.
Laboratory	M411A

Student Name	Chetan katkar
Roll Number	23102A0024
Grade and Subject Teacher's Signature	

Experiment Number	04
Experiment Title	Image Processing and Handling in R Programming
Objectives (Skill Set / Knowledge Tested / Imparted)	To understand image manipulation, read multiple image formats into R using appropriate libraries (such as EImage), explore image properties (color mode, storage mode, dimensions, frames), inspect raw image data matrices, generate image intensity histograms, resize images systematically in batches, and maintain project version control using Git and GitHub.
Description	In this practical, an image processing project was implemented in RStudio using image handling libraries. Multiple image files (p1.jpg through p6.jpg and c1.jpg through c6.jpg) were placed in the working directory and systematically imported into a list structure using a for loop. The loaded images were explored to check structural metadata including colorMode (Color), storage.mode (double), dimensions (275 × 183 × 3), total frames, and raw imageData numerical matrices. Individual images were displayed within the R environment, and summary statistics and pixel intensity distribution histograms were generated to analyze color channel distributions across pixel intensities (0.0 to 1.0). Finally, batch processing was performed to iteratively resize all images to uniform dimensions (28 × 28 pixels). The complete project, source code, and assets were managed and tracked

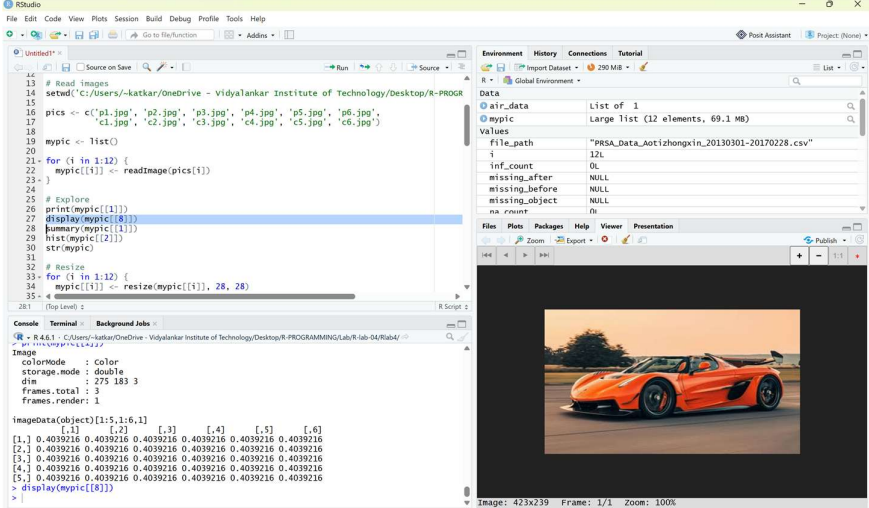
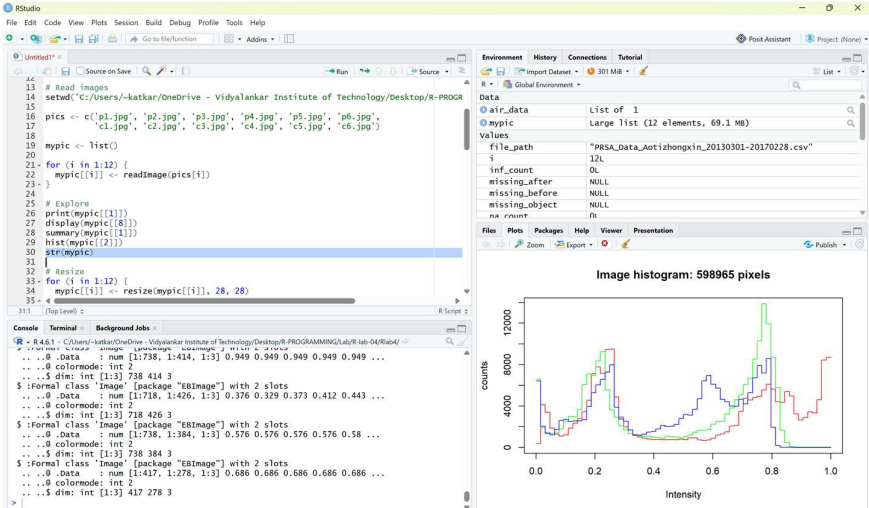
	using Git version control and published with comprehensive documentation to GitHub.
Output	<pre>> print(mypic[[1]]) Image colorMode : Color storage.mode : double dim : 275 183 3 frames.total : 3 frames.render : 1 imageData(object)[1:5,1:6,1] [,1] [,2] [,3] [,4] [,5] [,6] [1,] 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 [2,] 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 [3,] 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 [4,] 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 [5,] 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216</pre>  <pre>## R: R4.1.1, C:\Users\katar\OneDrive - Vidyalankar Institute of Technology\Desktop\R-PROGRAMMING\Lab-04\RSud\ > print(mypic[[1]]) Image colorMode : Color storage.mode : double dim : 275 183 3 frames.total : 3 frames.render : 1 imageData(object)[1:5,1:6,1] [,1] [,2] [,3] [,4] [,5] [,6] [1,] 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 [2,] 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 [3,] 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 [4,] 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 [5,] 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 > display(mypic[[8]]) ></pre>  <pre>## R: R4.1.1, C:\Users\katar\OneDrive - Vidyalankar Institute of Technology\Desktop\R-PROGRAMMING\Lab-04\RSud\ > print(mypic[[1]]) Image colorMode : Color storage.mode : double dim : 275 183 3 frames.total : 3 frames.render : 1 imageData(object)[1:5,1:6,1] [,1] [,2] [,3] [,4] [,5] [,6] [1,] 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 [2,] 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 [3,] 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 [4,] 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 [5,] 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 0.4039216 > display(mypic[[8]]) ></pre>
Conclusion	Thus, image loading, structural exploration, visualization, histogram analysis, and batch resizing were successfully implemented using R Programming. The practical provided hands-on experience in working with multidimensional image arrays, analyzing pixel intensity distributions, automated batch

	image transformations, and integrating standard software engineering practices such as Git version control and GitHub repository management for R-based projects.
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